

SAMBP Technical Report TR-36/2026

Auto-Reclosing Strategy for Inverter-Based Resource Lines

PhD Thesis Supporting Report | 2026

Abstract

Auto-reclosing (AR) on IBR-fed transmission lines creates a fundamental conflict: the arc deionisation dead time always exceeds the memory-polarisation hold time ($T_{\text{mem}} = 100$ ms), so the protection elements relied upon for the first trip (Zone 1 quadrilateral, Z2/POTT) are unavailable when the breaker recloses onto a permanent fault. This report quantifies the tension using a current-dependent arc model ($T_{\text{arc}} \propto \sqrt{k_{\text{ibr}}}$) and finds that IBR-limited fault current actually reduces arc deionisation time relative to synchronous-machine faults: at $k_{\text{ibr}} = 0.15$, $T_{\text{dead}} = 205$ ms versus 450 ms for a SG-only line. All dead times remain within the IBR under-voltage ride-through window of 625 ms. However, T_{dead} always exceeds T_{mem} , so the second-trip path on reclose-onto-permanent-fault relies exclusively on 87L (primary, 20 ms) and 67Q/67N (backup, 30 ms), both of which are memory-independent. Three-phase, single-shot AR with synchrocheck is recommended; single-phase AR is inappropriate due to sustained negative-sequence injection into IBR current controllers during the open-phase interval.

Motivation

Approximately 80% of transmission-line faults are transient (lightning overvoltages, temporary conductor contact) and can be cleared by opening the circuit breaker and reclosing after the arc self-extinguishes. AR therefore recovers most fault events without a prolonged outage.

For IBR-fed lines the AR process has two complicating features not present on conventional SG lines:

1. **Memory polarisation expiry:** the first trip correctly uses Zone 1 or Z2/POTT (TR-30, TR-33 [? ?]), which depend on memory polarisation valid within $T_{\text{mem}} = 100$ ms. The CB dead time ($> T_{\text{mem}}$) means memory has expired before the breaker closes. If the fault is permanent, the second trip must use only memory-independent elements.
2. **IBR current limiting during fault:** the IBR contributes reduced fault current ($I_{\text{IBR}} = k_{\text{ibr}} \cdot I_{\text{SG}}$), which lowers arc energy and — usefully — shortens the arc deionisation time.

Arc Deionisation Model

The arc deionisation time depends on the ionisation level in the arc channel, which is proportional to the fault current. Higher current injects more energy into the arc, producing a more ionised gas column that takes longer to deionise [?]:

$$T_{\text{arc}}(I_{\text{fault}}) = T_{\text{arc},0} \sqrt{\frac{I_{\text{fault}}}{I_{\text{base}}}}, \quad (1)$$

where $T_{\text{arc},0} = 400$ ms at the synchronous-generation base fault current $I_{\text{base}} = 10$ kA (230 kV line, ANSI/IEEE C37.04 reference). For an IBR line, $I_{\text{fault}} = k_{\text{ibr}} \cdot I_{\text{base}}$, so:

$$T_{\text{arc,IBR}} = T_{\text{arc},0} \sqrt{k_{\text{ibr}}} < T_{\text{arc,SG}} = T_{\text{arc},0}. \quad (2)$$

The minimum required dead time includes a 50 ms safety margin:

$$T_{\text{dead}}(k_{\text{ibr}}) = T_{\text{arc},0} \sqrt{k_{\text{ibr}}} + 50 \text{ ms}. \quad (3)$$

Study A — Dead Time vs IBR Penetration

Figure 1 plots T_{arc} and T_{dead} versus k_{ibr} , with the memory expiry threshold and IBR ride-through limit overlaid.

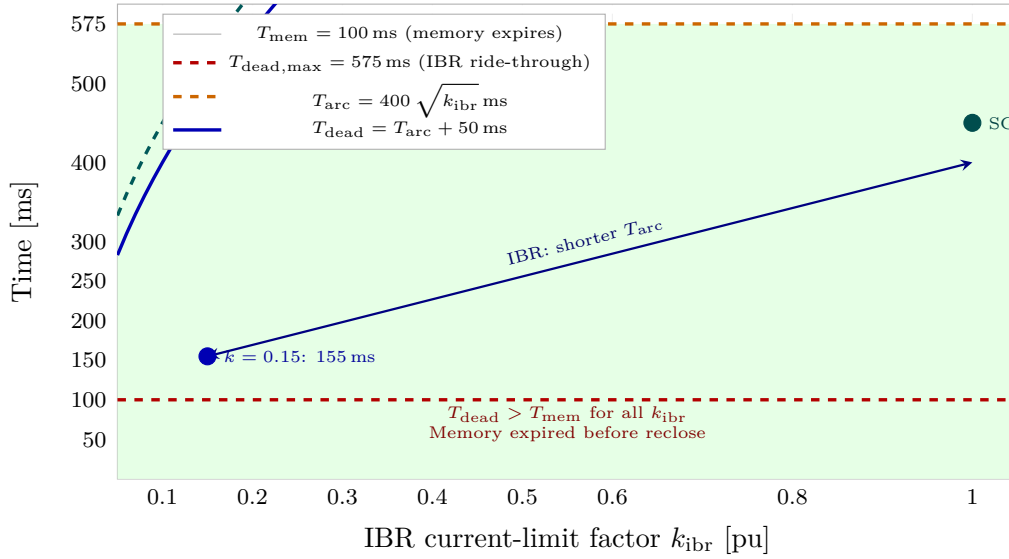


Figure 1: Arc deionisation and dead time vs k_{ibr} . The IBR advantage (lower fault current $\Rightarrow$ shorter T_{arc}) is clear: T_{dead} drops from 450 ms at SG level to 176 ms at $k_{\text{ibr}} = 0.10$. All dead times are below the IBR ride-through limit of 575 ms (green shaded region), confirming AR feasibility for $k_{\text{ibr}} \geq 0.10$. However, all dead times exceed $T_{\text{mem}} = 100$ ms (red dashed), so memory is always expired before reclose.

Representative dead times:

k_{ibr}	I_{fault} [kA]	T_{arc} [ms]	T_{dead} [ms]	≤ 575 ms?	IBR connected?
0.10	1.0	127	177	Yes	Yes
0.15	1.5	155	205	Yes	Yes
0.20	2.0	179	229	Yes	Yes
0.50	5.0	283	333	Yes	Yes
1.00	10.0	400	450	Yes	Yes

Study B — Protection Availability at Second Trip

When the CB recloses onto a permanent fault, the protection relay must trip a second time. At this point:

- T_{mem} expired: Zone 1 and Z2/POTT lack a valid polarising reference $\Rightarrow$ **unavailable**.
- 87L differential: operates on instantaneous current differential — no polarisation memory required $\Rightarrow$ **available, 20 ms**.
- 67Q/67N: operate on sequence current direction — no memory required $\Rightarrow$ **available, 30 ms**.

α	k_{ibr}	Z1 avail.?	Z2/POTT?	67N/67Q?	87L?
0.50	0.10	No	No	Yes	Yes
0.50	0.15	No	No	Yes	Yes
0.50	0.20	No	No	Yes	Yes
0.90	0.15	No	No	Yes	Yes

87L is the definitive second-trip element. This reinforces the TR-35 conclusion [?] that 87L is indispensable: it functions both as a first-trip co-primary and as the only distance-independent clearance path on reclose-onto-permanent-fault.

Study C — IBR Ride-Through Feasibility

The total voltage-depression duration seen by the IBR is:

$$\Delta t_{\text{UVT}} = t_{\text{first_trip}} + t_{\text{CB_open}} + T_{\text{dead}} = 32 + 50 + T_{\text{dead}} \text{ ms.} \quad (4)$$

Grid code (IEC 61727 [?]) requires IBR to ride through low voltage for at least 625 ms. With the values in Section 3, $\Delta t_{\text{UVT}} \leq 414 \text{ ms}$ for $k_{ibr} \leq 0.50$ and $\leq 532 \text{ ms}$ for $k_{ibr} = 1.00$ — all within the 625 ms limit. The IBR remains connected through the full AR cycle and resumes full output within approximately 100 ms of successful reclose.

Study D — Single-Phase vs Three-Phase AR

Single-phase (1PH) AR is common on EHV networks with SG infeed, as it maintains power transfer on the two healthy phases. For IBR-fed lines it is inadvisable:

Criterion	1PH AR	3PH AR
Transient fault clearance	Adequate	Adequate
Negative-seq during open phase	V_2 sustained for T_{dead}	None
IBR I_2 stress	Continuous injection risk (TR-32)	Eliminated
Reclose onto permanent SLG	Re-energises faulted phase	Trips all phases cleanly
Synchrocheck	2-phase reference simpler	Full 3-phase check needed

The sustained V_2 during the 1PH open interval drives the IBR 67Q negative- sequence directional element (TR-32 [?]) into a continuous forward assertion, potentially stressing the IBR's current-control loops for 200–450 ms. IBR manufacturers do not typically certify this operating mode. Three-phase AR avoids all asymmetric-loading effects.

Study E — Complete AR Scheme

Figure 2 shows the AR state machine; Figure 3 shows the event timeline for $k_{ibr} = 0.15$. The recommended AR scheme for IBR-fed lines:

1. **Mode:** 3-phase, single-shot.
2. **Dead time:** $T_{\text{dead}} = T_{\text{arc},0} \sqrt{k_{ibr}} + 50 \text{ ms}$, adaptive to IBR penetration level.
3. **Synchrocheck:** angle difference $|\Delta\delta| \leq 20^\circ$, voltage difference $|\Delta V| \leq 5\%$ before reclose permitted.
4. **Second-trip elements:** 87L (primary) and 67Q/67N (backup); Zone 1 and Z2/POTT inhibited until fresh-fault memory is established.
5. **Lock-out:** after one failed reclose; manual restoration required.

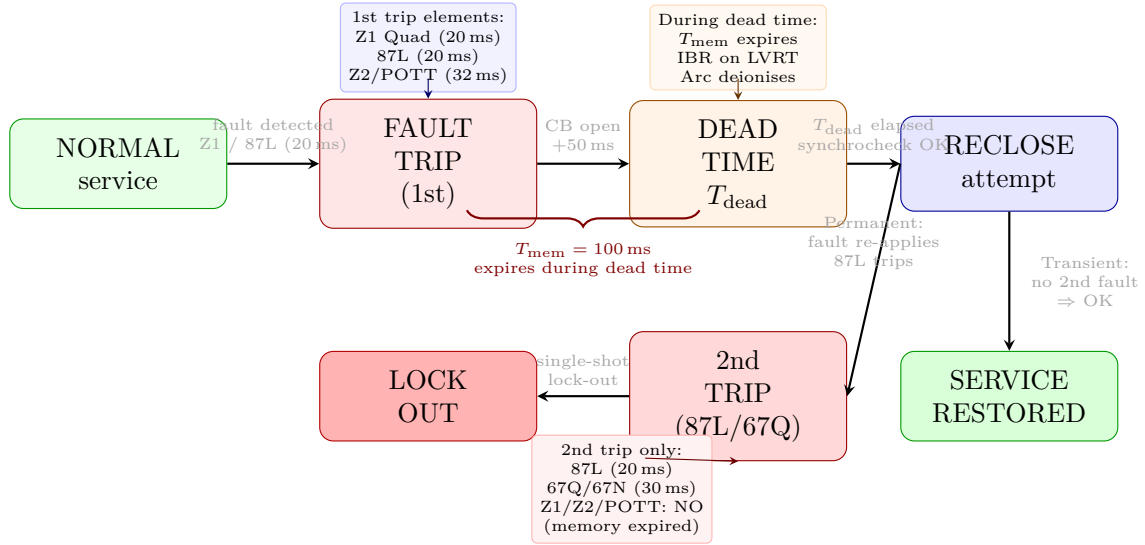


Figure 2: AR state machine. First trip uses memory-polarised elements (Z1 / Z2/POTT / 87L). During the dead time, T_{mem} expires. On reclose: transient fault $\Rightarrow$ service restored; permanent fault $\Rightarrow$ second trip by 87L/67Q (memory-independent), then lock-out. Single-shot scheme: no second reclose attempt.

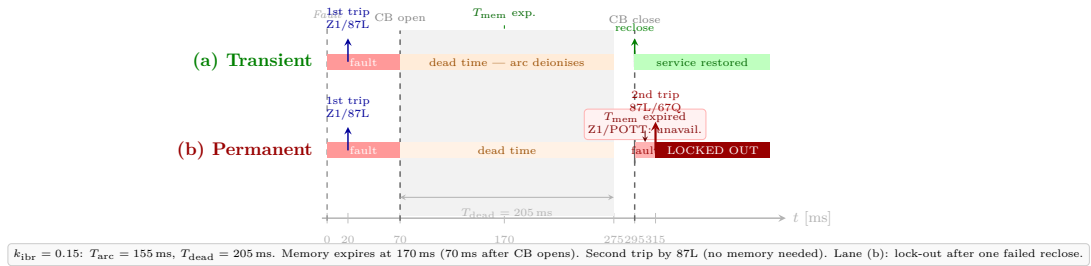


Figure 3: AR event timeline for $k_{\text{ibr}} = 0.15$ ($T_{\text{dead}} = 205$ ms). Lane (a): transient fault — arc quenches during dead time, reclose at 275 ms succeeds. Lane (b): permanent fault — reclose at 275 ms re-applies fault, 87L clears at 315 ms (memory long expired), relay locks out. Note the memory expiry at 170 ms (70 ms after CB opens) — well before the reclose.

Summary

1. **Dead time (Study A):** IBR-limited fault current reduces T_{arc} relative to SG faults: 127 ms at $k = 0.10$ versus 400 ms at SG level. Dead times are 177–450 ms — all feasible within the IBR 625 ms ride-through limit.
2. **Memory expiry (Studies A, B):** T_{dead} invariably exceeds $T_{\text{mem}} = 100$ ms. Zone 1 and Z2/POTT are unavailable at second trip. 87L (20 ms) and 67Q/67N (30 ms) are the sole reliable clearance paths after reclose.
3. **IBR ride-through (Study C):** The full AR cycle ($t_{\text{trip}} + t_{\text{CB}} + T_{\text{dead}}$) stays within 625 ms for all k_{ibr} ; IBR remains connected.
4. **3PH AR (Study D):** Three-phase reclosing is preferred; single-phase AR imposes sustained negative-sequence stress on IBR current-control loops during the open-phase interval.
5. **Scheme (Study E):** Single-shot 3PH AR with adaptive dead time, synchrocheck, and 87L/67Q as second-trip elements completes the SAMBP auto-reclosing design.

References

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- —, “SAMB P TR-33/2026: Permissive overreaching transfer trip (POTT) scheme for IBR lines,” PhD Thesis Supporting Report, Tech. Rep., 2026.
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- *IEC 61727: Photovoltaic (PV) Systems — Characteristics of the Utility Interface*, International Electrotechnical Commission Std., 2004.